Al-Idrisi's World: The Map Made for Roger II of Sicily in 1154, Set Against a Modern Map

431 towns on Konrad Miller's reconstruction, measured for direction, distance, climate and scale, and compared with the Hereford Mappa Mundi

Tristan Hotham

Independent researcher
ORCID: 0000-0003-4931-9349
DOI: https://doi.org/10.5281/zenodo.23882607

2026

Abstract

Al-Idrisi's world map, finished in Palermo in 1154 for Roger II of Sicily, is drawn with south at the top on a grid of seven climates and ten sections. I located 431 legible town labels on Konrad Miller's 1928 reconstruction of the map, identified each with a modern place, checked the positions against three gazetteers, and measured how the map, turned north up, distorts the world. Directions are well kept: across 92,193 pairs of towns at least 100 km apart the median error in bearing is 10 degrees and 99% lie within 45 degrees. The best affine fit of the whole map puts the median town 239 km from its real position. Locally the map is more faithful: the median distance between a town and the position predicted from its eight nearest neighbours is 55 km, rising from 44 km around the Mediterranean to 71 km in northern Europe and 92 km in Asia beyond the Euphrates. Sicily is both the most faithfully drawn region (21 km) and the most enlarged (1.42 times the map's average scale). The climates keep the order of latitude but not its spacing: the boundaries implied by the towns fall at about 18.0, 28.6, 35.8, 40.4, 47.3 and 53.2 degrees north. 63 route distances from Dozy and de Goeje's 1866 edition give a median of 1.42 km of straight line per stated mile and about 36 km per day stage. Measured exactly as in my study of the Hereford Mappa Mundi (about 1300), al-Idrisi's map explains 0.94 of the spread of real positions about Jerusalem against 0.84, with a median miss of 330 km against 516 km.

Keywords: medieval cartography; Islamic geography; al-Idrisi; Tabula Rogeriana; Roger II; map distortion; climates; historical gazetteers; open data

1 Introduction

Abu Abd Allah Muhammad al-Idrisi, born in Ceuta, worked at the court of Roger II, the Norman king of Sicily, and in 1154 completed for him a geography, the *Nuzhat al-mushtaq*, often called the Book of Roger, together with a world map. The work divides the inhabited world into the seven climates of Greek and Islamic geography, bands of latitude from the equator northward, and each climate into ten sections from west to east; each of the seventy sections has its own description and its own sheet. Joined, the sheets form a rectangular map of the world with south at the top. The original is lost; the map is known from manuscript copies of the section maps, which Konrad Miller assembled into a single reconstruction in 1928 (Miller, 1928).

This paper sets that reconstruction against a modern map. It asks how well the map keeps direction and distance; where it is most and least faithful; whether its climates follow latitude; how it treats the Nile and Britain; how its book's stated distances compare with real ones; and how it compares, on the same measures, with the Hereford Mappa Mundi, a Christian world map drawn about 146 years later, which I analysed in an earlier study (Hotham, 2026). It accompanies

an interactive comparison (https://maps.tristanhotham.com/idrisi/).

2 Data

The map. Miller's reconstruction (Miller, 1928), from Wikimedia Commons (public domain), scanned at 12,999 by 5,958 pixels. Miller redrew the map from the manuscripts and transcribed every name into Latin letters; all positions in this study are therefore positions on Miller's sheet, not on a medieval manuscript.

The text. Dozy and de Goeje (1866) edited and translated the parts of the Book of Roger on North Africa and Spain. The Internet Archive's text of that edition (public domain) supplies the route distances.

Gazetteers. al-Thurayya (Seydi and Romanov, 2026), a gazetteer of the places and routes of the early Islamic world with modern coordinates, derived from Cornu's atlas (CC BY 4.0); Pleiades (Pleiades, 2026), version 4.1 (CC BY 3.0); and Natural Earth countries and populated places (Natural Earth, 2026).

3 Methods

3.1 Reading and locating towns

Miller drew settlements as small brown discs with spokes, each beside its transcribed name. I read the labels section by section at full resolution and recorded, for each town I could read and identify with confidence, the pixel of its disc and my confidence (362 read clearly, 69 with some doubt because the label is crowded, faded or ambiguous). I left out rivers, mountains, regions and peoples, and every label I could not identify. To make positions reproducible I detected the discs automatically, first by shape (a colour mask, a morphological opening and a test for round blobs of the right size) and then by matching a template built from the clean discs; 5,329 candidate discs were found, and 386 of my readings (90%) were snapped to a disc within 20 pixels, a median 1.9 pixels away. Readings left unsnapped were checked by eye on close-up images, and eleven positions were corrected in this way.

3.2 Modern positions

I gave each identification a modern position and checked it against the three gazetteers by name: al-Thurayya by Miller's transliteration and by the identification, Pleiades by attested and romanised names, Natural Earth by modern names. 365 of the 431 towns (85%) matched a gazetteer entry of the same name within 150 km, at a median distance of 0.7 km from my position (178 in al-Thurayya, 264 in Pleiades, 196 in Natural Earth); every match more than 30 km away was reviewed. The towns not matched are small places the gazetteers do not list. Each town was assigned to one of 18 regions (Table 1).

3.3 Measures

All measures use the map turned through 180 degrees, so that north is up. The *global fit* is the affine map (least squares) from map position to longitude and latitude; the *residual* of a town is

the great circle distance between its real position and its fitted position. The *local error* of a town is obtained by fitting an affine map from map position to an azimuthal equidistant projection centred on the town, using only its eight nearest neighbours by true distance, and measuring how far that fit puts the town from its real position; it measures how well the map keeps the shape of each neighbourhood, regardless of distortion at the scale of the whole map. For each region with at least five towns I also computed, over all pairs of towns in the region at least 100 km apart, the median absolute difference between the bearing on the map and the true initial bearing, and the median ratio of map distance to true distance relative to the same median over all pairs on the map (the *stretch*, as in the Hereford study). Because the map is a rectangular grid, stretch was also computed for a perfect rectangular grid with the fitted ratio of degrees of longitude to degrees of latitude, to separate the effect of the grid from the mapmaker's choices.

The climate of each town is read from Miller's red climate lines, located automatically in each of the ten sections. For each pair of neighbouring climates, the *implied boundary* is the latitude that best separates the towns of the two climates. For the comparison with Hereford I repeated that study's method exactly: the similarity transform (rotation, uniform scale and shift) that best lays the map positions on an azimuthal equidistant projection centred on Jerusalem, and bearings from Jerusalem on the map and on the globe.

3.4 Route distances

From Dozy and de Goeje (1866) I took 63 distances stated between two places I could identify, over land or along a coast: 37 in miles and 30 in day stages (some give both). Each was checked against the line of the scan's text it comes from, which is recorded in the data. I compared each with the great circle distance between the places.

4 Results

4.1 The towns and the whole-map fit

Figure 1 shows the 431 towns on the map and on a modern map. The global affine fit explains 0.98 of the variance in longitude and 0.90 in latitude. It draws a degree of longitude 0.93 times as long as a degree of latitude, the ratio of a rectangular grid true to scale at about 21 degrees north. The median residual is 239 km, 54% of towns are within 250 km of their fitted position and 88% within 500 km (Figure 2).

4.2 Direction and local shape

Directions are well kept. Across all 92,193 pairs of towns at least 100 km apart, the median bearing error is 10 degrees and 99% are within 45 degrees; the rank correlation between map and true distance is 0.95. The median local error is 55 km, with 76% of towns within 100 km and 91% within 200 km. Table 1 and Figure 3 give the figures by region. The most faithful regions are Sicily (21 km), Syria and Palestine (26 km), Morocco and Algeria and Iberia (37 km each), and Italy (38 km); the least faithful are Arabia (156 km), central and eastern Europe (212 km) and Sind and India (247 km). Grouped, the Mediterranean regions have a median local error of 44 km (275 towns), northern Europe 71 km (50) and Asia beyond the Euphrates 92 km (105).



Figure 1: The located towns on Miller's reconstruction, turned north up (top), and on a modern map (bottom), coloured by region.

4.3 Scale and Sicily

Relative to the map's average scale, Sicily is drawn largest (1.42), followed by Greece and the Balkans (1.38), England and Italy (1.25) and Syria and Palestine (1.25); Morocco and Algeria (0.68) and Arabia (0.59) are drawn smallest. A perfect rectangular grid with the map's proportions would leave Sicily at 0.99, so its enlargement is not an artefact of the grid. The local fit for Sicily is rotated by about 31 degrees, which explains why its bearing error (28 degrees) is among the highest even though its local error is the lowest (Figure 4).

4.4 The climates

The median latitude of the towns rises without a break from climate to climate (Table 2, Figure 5). Every climate is drawn about the same height (a mean of 754 pixels), but the implied boundaries fall at 18.0, 28.6, 35.8, 40.4, 47.3 and 53.2 degrees north: the second climate spans 10.6 degrees and

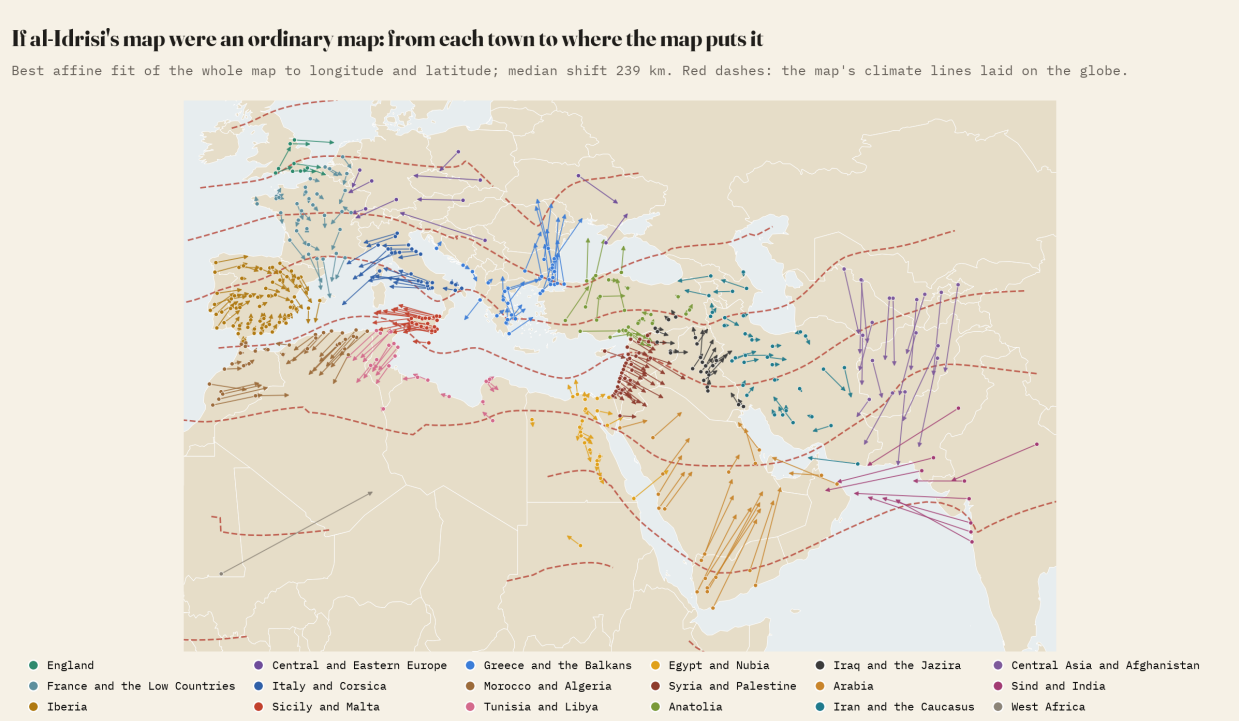


Figure 2: From each town to its position under the best affine fit of the whole map, with the map's climate lines carried onto the globe.

Table 1: Located towns by region: number, median local error (km), median residual of the global affine fit (km), median bearing error between pairs (degrees), stretch relative to the map's average scale, and the stretch a perfect rectangular grid would give. Blank: fewer than five towns.

Region	Towns	Local	Global	Bearing	Stretch	Grid
England	8	63	175	39	1.25	0.98
France and the Low Countries	32	57	143	18	1.18	1.00
Iberia	64	37	140	23	0.84	0.99
Central and Eastern Europe	10	212	379	15	0.92	1.16
Italy and Corsica	24	38	369	25	1.25	1.01
Sicily and Malta	20	21	362	28	1.42	0.99
Greece and the Balkans	32	76	315	20	1.38	0.99
Morocco and Algeria	33	37	353	19	0.68	0.99
Tunisia and Libya	20	50	253	18	0.93	0.95
Egypt and Nubia	25	61	156	17	1.13	0.89
Syria and Palestine	33	26	249	20	1.25	0.90
Anatolia	24	87	253	20	1.04	1.00
Iraq and the Jazira	23	54	204	22	0.87	0.93
Arabia	21	156	759	17	0.59	0.88
Iran and the Caucasus	34	81	128	17	0.86	0.92
Central Asia and Afghanistan	18	137	669	24	0.84	0.94
Sind and India	9	247	1,112	39	0.73	0.88
West Africa	1	664	1,921			

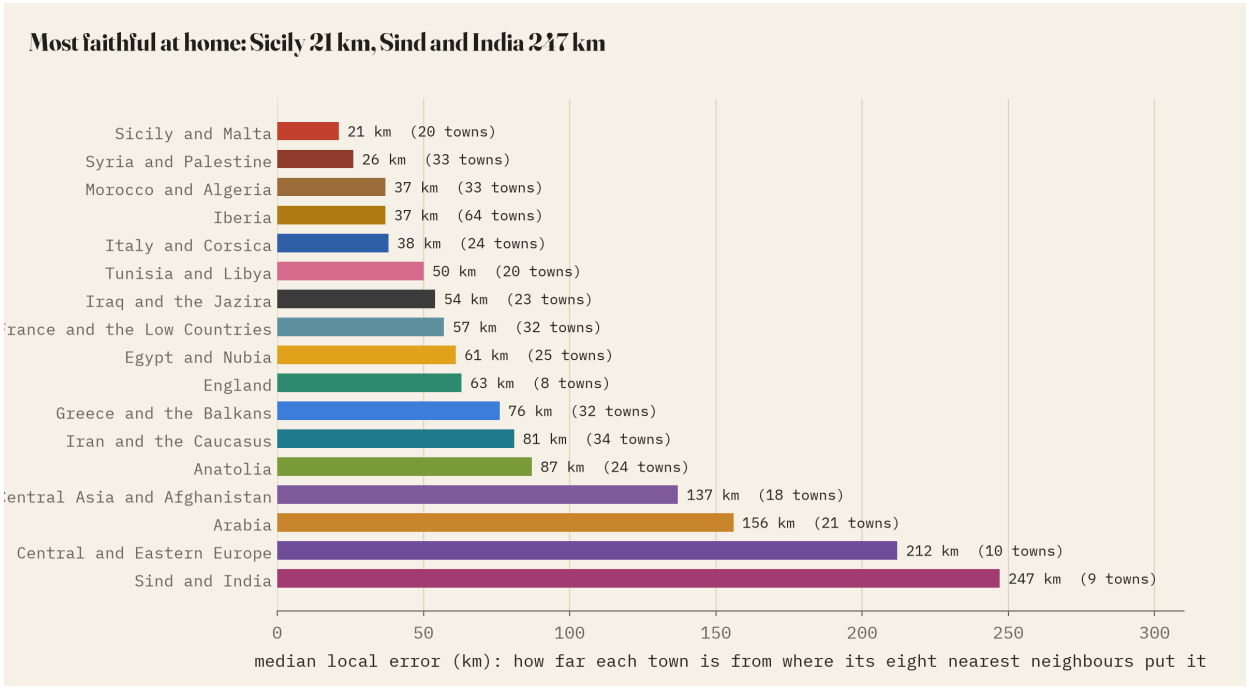


Figure 3: Median local error by region: the distance from each town to the position predicted by an affine fit to its eight nearest neighbours.



Figure 4: Sicily on Miller's reconstruction (turned north up) and today.

Table 2: Towns by climate: number, minimum, median and maximum true latitude (degrees north), and the implied boundary with the next climate.

Climate	Towns	Min	Median	Max	Boundary
I	10	12.8	15.6	24.1	18.0
II	36	14.8	25.3	34.6	28.6
III	108	25.4	32.8	37.0	35.8
IV	149	32.0	37.6	43.2	40.4
V	80	37.8	41.3	47.3	47.3
VI	45	41.0	48.4	52.5	53.2
VII	3	51.1	53.2	53.6	

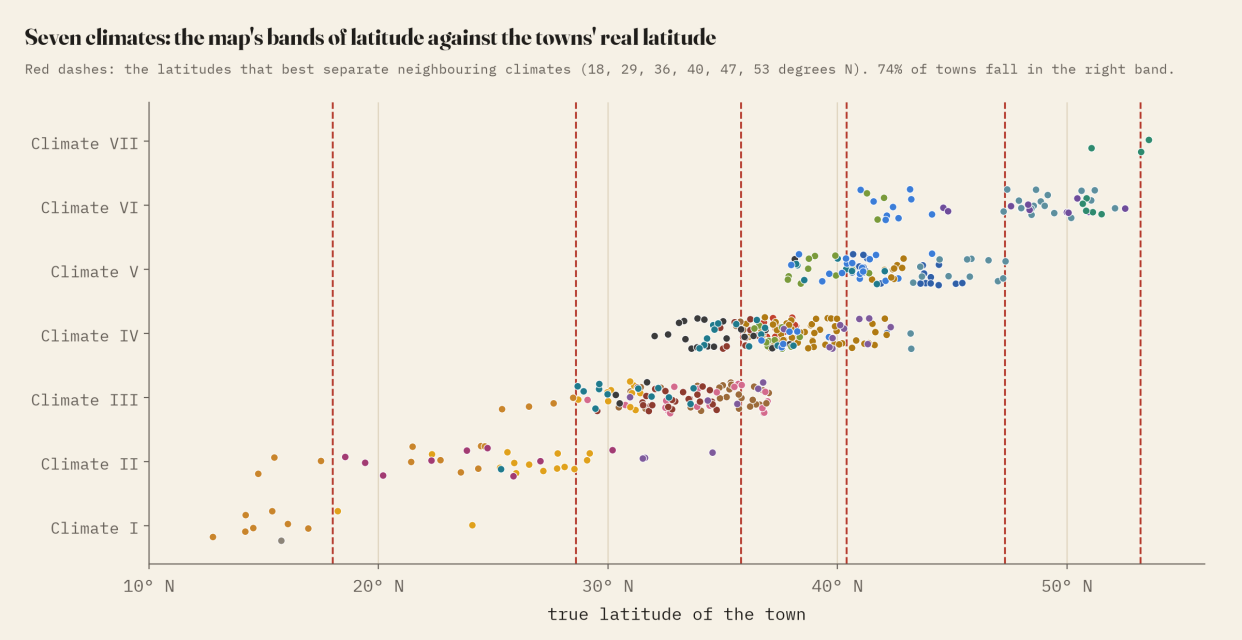


Figure 5: True latitude of the towns in each climate. Dashed lines: implied boundaries.

the third to sixth an average of 5.8. With these boundaries 74% of towns lie in the climate their latitude would give and all lie within one. The first and seventh climates are thinly sampled (10 and 3 towns).

4.5 The Nile

The text places the source of the Nile in the Mountains of the Moon, "whose beginning is sixteen degrees beyond the equinoctial line" (Dozy and de Goeje, 1866), and quotes Qudama that the river runs 5,654 miles from source to sea. On Miller's sheet the mountains sit only about 130 pixels beyond the equator line, about 0.17 of the height of a climate, so the map does not show sixteen degrees. At 1.92 to 2.0 km to the Arab mile, 5,654 miles is 10,900 to 11,300 km, against about 6,650 km for the real river. Along the river itself the map is good: under the global fit Aswan falls at 23.3 degrees north (really 24.1) and Alexandria at 32.5 (really 31.2), and Egypt's towns span 13.7 degrees of fitted latitude against 13.2 in reality.

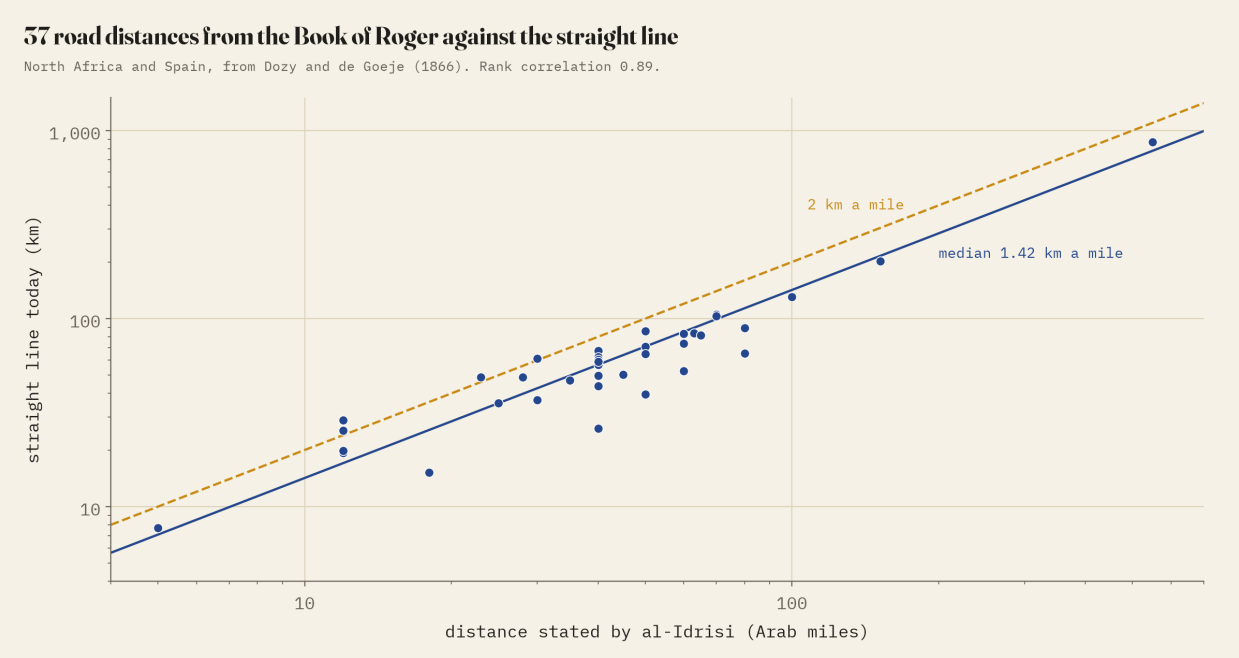


Figure 6: Distances stated in the Book of Roger (North Africa and Spain) against the great circle distance between the two places.

4.6 Britain

I could read 8 English towns, too few for firm conclusions. They straddle the boundary between the sixth and seventh climates. Under the global fit their median latitude error is within a tenth of a degree, while they are displaced 2.3 degrees of longitude to the east, and the local fit is rotated about 37 degrees clockwise: Paris, 148 degrees from London in reality, is drawn at 191 degrees. Measured from Jerusalem, the English towns point the right way to within a median 12 degrees, against 8 for the British places on the Hereford map.

4.7 Route distances

The stated miles rise with the straight line distance (Spearman 0.89; Figure 6), at a median of 1.42 km per mile (interquartile range 1.23 to 1.58). If the Arab mile is taken as 1.92 to 2.0 km, the stated road distances are 1.35 to 1.41 times the straight line, plausible for roads; alternatively the miles of these itineraries were shorter. A day stage is a median 36 km of straight line (interquartile range 30 to 43; Spearman 0.83), and in the 4 routes that give both, a day is about 25 miles. For the 18 routes in miles with both ends located on the map, map distance correlates more closely with the stated miles (Spearman 0.74) than with the true distance (0.48), consistent with a map compiled from itineraries, though the sample is small.

4.8 Comparison with the Hereford Mappa Mundi

On the Hereford study's measures (Figure 7), the similarity fit about Jerusalem explains 0.94 of the spread of real positions on al-Idrisi's map against 0.84 on Hereford; the median displacement is 330 km against 516 km; the median bearing error from Jerusalem is 13 degrees against 19; and 99% of places lie within 45 degrees against 81%. Hereford draws the Holy Land at 4.5 times its average scale; al-Idrisi draws Syria and Palestine at 1.25.

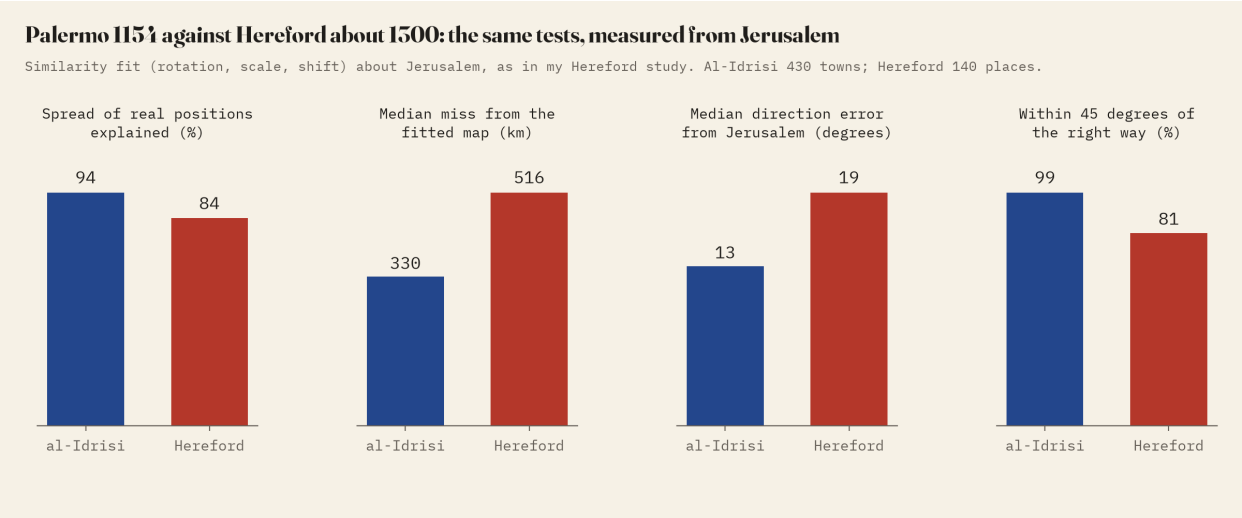


Figure 7: The same tests on al-Idrisi's map and on the Hereford Mappa Mundi.

5 Limitations

First, the map measured is Miller's reconstruction, which joins and redraws the section maps of the manuscripts; where his work departs from them, so do these measurements. Second, the sample is the legible, identifiable towns, about 431 of a much larger number of names; it favours well documented places, and it is thin in India and Central Asia, eastern Europe and Africa south of the Sahara (9, 10 and 1 towns). Third, the identifications and modern positions are mine; 85% were confirmed in a gazetteer, and the rest are small places whose identification rests on the name and its position among its neighbours. Fourth, with clear readings only (362 towns) the median local error is 52 km and the median global residual 235 km, close to the full sample's. Fifth, the route distances come from one edition covering only North Africa and Spain, read from OCR and checked line by line, and the length of the Arab mile is uncertain. Sixth, the Hereford comparison sets a sample of medieval towns against a sample dominated by classical names; it compares the maps as sources of positions, not their makers' purposes.

6 Conclusion

Turned north up, al-Idrisi's map is a recognisable map of the Old World. It keeps directions between towns to a median of 10 degrees and keeps the shape of neighbourhoods best where its informants were most numerous: around the Mediterranean, and above all in Sicily, which it also draws larger than anywhere else. Its climates order the world by latitude while compressing the south and spreading out the populous middle latitudes. On the measures I applied to the Hereford Mappa Mundi, it is the more accurate map, as befits a geographer's survey compiled from travellers rather than a picture of Scripture and history; but both spend their space most generously on home.

Data and code availability

The interactive comparison is at https://maps.tristanhotham.com/idrisi/. Data and code are at ht
tps://github.com/Tristan-Hotham/idrisi and archived with this paper at https://doi.org/10.5281/zenodo.23882

Derived data are released under CC BY 4.0 and code under the MIT licence. Miller's reconstruction and the 1866 edition are in the public domain; al-Thurayya data are CC BY 4.0 and Pleiades data CC BY 3.0.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

References

Dozy, R. and de Goeje, M. J. (1866) *Description de l'Afrique et de l'Espagne par Edrisi*. Leiden: Brill. Internet Archive scan, public domain. https://archive.org/details/bub_gb_6EMNNqPF2zAC

Hotham, T. (2026) The world according to the Hereford Mappa Mundi: a medieval map of the world set against a modern one. Zenodo. https://doi.org/10.5281/zenodo.23878612

Miller, K. (1928) *Charta Rogeriana: Weltkarte des Idrisi vom Jahr 1154 n. Ch.* Stuttgart. Wikimedia Commons, public domain. https://commons.wikimedia.org/wiki/File:Tabula_Rogeriana_1929_copy_by_Konrad_Miller.jpg

Natural Earth (2026) Admin 0 countries and populated places, 1:50m and 1:10m. Public domain. https://www.naturalearthdata.com/

Bagnall, R. et al., eds. (2026) Pleiades: a gazetteer of past places, version 4.1. CC BY 3.0. https://pleiades.stoa.org/

Seydi, M. and Romanov, M. (2026) al-Thurayya Gazetteer, places and routes of the early Islamic world, from G. Cornu (1983) *Atlas du monde arabo-islamique*. CC BY 4.0. https://github.com/althurayya/althurayya.github.io

A The route distances

Table 3: Distances stated by al-Idrisi in Dozy and de Goeje (1866): miles, day stages, great circle distance today (km) and the line of the Internet Archive text.

From	To	Miles	Days	km	Line
Fez	Meknes	40		57	5062
Aghmat	Marrakesh	12		29	4623
Jijel	Bejaia	30		61	6297
Algiers	Tamenfoust	18		15	5859
Bizerte	Tabarka	70		105	7275
Monastir	Mahdia	30		37	8024
Tocra	Ptolemais (Tolmeita)	50		40	8513
Barca (al-Marj)	Ajdabiya	152	6	202	8392
Barca (al-Marj)	Alexandria	550	21	870	8394
Rosetta (Rashid)	Alexandria	60		53	10097
Cadiz	Tarifa	63		84	11186
Faro (Santa Maria)	Silves	28		49	11223
Guadalajara	Medinaceli	50		86	11809
Daroca	Zaragoza	60		74	11850
Calatayud	Zaragoza	50		71	11852
Zaragoza	Huesca	40		68	11881
Huesca	Lleida	70		103	11883
Mequinenza	Tortosa	50	2	65	11890
Tarragona	Barcelona	60		83	11913
Alzira	Xativa	12		19	11982
Valencia	Denia	65		82	11992
Valencia	Cullera	25		36	11994
Xativa	Bocairent	40		26	12026
Bocairent	Denia	40		63	12035
Orihuela	Murcia	12		20	12074
Cartagena	Murcia	40		44	12153
Lorca	Murcia	40		61	12299
Almunecar	Granada	40		50	12487
Granada	Loja	23		49	12868
Almunecar	Malaga	80		65	12872
Malaga	Granada	80		89	12890
Lucena	Cordoba	40		59	12992
Ecija	Cordoba	35		47	13012
Ecija	Carmona	45		50	13020
Jerez de la Frontera	Isla de Leon (San Fernando)	12		25	13049
Cordoba	Madinat al-Zahra	5		8	13475
Cordoba	Granada	100	4	131	13655
Sale	Fez		4	168	4995
Fez	Sijilmasa		13	317	5042
Tlemcen	Tahert (Tiaret)		4	246	5683
Tenes	Miliana		2	86	5798
M'Sila	Tobna		2	83	6116
Tobna	Bejaia		6	158	6130
Constantine	Bejaia		6	143	6295
Collo	Constantine		2	71	6406
Setif	Constantine		4	110	6418
Sfax	Gafsa		3	184	6911
Sfax	Mahdia		2	89	6934
Kairouan	Tunis		2	126	7097
Barca (al-Marj)	Awjila		10	380	8390
Fustat (Old Cairo)	Oxyrhynchus (al-Bahnasa)		7	172	10130
Fustat (Old Cairo)	Aswan		26	678	3886
Algeciras	Seville		5	148	11025

From	To	Miles	Days	km	Line
Silves	Badajoz		3	228	11252
Evora	Badajoz		2	88	11332
Badajoz	Seville		6	187	11356
Badajoz	Cordoba		6	220	11359
Merida	Alcantara		2	101	11451
Santarem	Badajoz		4	153	11631
Valencia	Zaragoza		9	246	11969
Murcia	Valencia		5	178	12180
Murcia	Almeria		5	174	12182
Cordoba	Almeria		8	236	13489

B The located towns

Table 4: The 431 located towns: identification, Miller's transcription, region, climate, my confidence in the reading, and local error (km).

Town	On the map	Region	Clim.	Reading	Local
Adana	adana	Anatolia	IV	high	81
Ahlat	halat	Anatolia	V	high	88
Amasra (Amastris)	samastru	Anatolia	VI	high	52
Amasya	amasia	Anatolia	V	high	137
Amorium	amuria	Anatolia	V	high	62
Anazarbus (Ayn Zarba)	ain zarba	Anatolia	IV	high	77
Ankara	ankira	Anatolia	V	medium	255
Antalya	antalia	Anatolia	IV	high	263
Bitlis	bidlis	Anatolia	V	high	53
Erzurum (Qaliqala)	kalikala	Anatolia	V	high	166
Iskenderun	iskanderuna	Anatolia	IV	high	93
Kahramanmaras	maras	Anatolia	IV	high	93
Karadeniz Eregli (Heraclea Pontica)	heraklia	Anatolia	VI	high	67
Kastamonu	kastamuni	Anatolia	V	high	159
Kayseri	kaisaria	Anatolia	V	high	233
Konya	kunia	Anatolia	V	high	203
Laodicea on the Lycus	ladikia	Anatolia	V	medium	152
Malatya	malatia	Anatolia	IV	high	87
Mopsuestia (al-Massisa)	al masisa	Anatolia	IV	high	71
Samosata (Samsat)	samsat	Anatolia	IV	high	38
Silifke (Seleucia)	selukia	Anatolia	IV	high	60
Sinop	sinuboli	Anatolia	VI	high	126
Tarsus	tarsus	Anatolia	IV	high	65
al-Hadath	al hadit	Anatolia	IV	high	57
Aden	aden	Arabia	I	high	156
Dhamar	damar	Arabia	I	high	134
Hajr al-Yamama (Riyadh)	hagar al iemama	Arabia	II	medium	165
Jeddah	gidda	Arabia	II	high	21
Madyan (al-Bad)	madjan	Arabia	III	medium	138
Marib	marib	Arabia	II	high	181
Mecca	mekka	Arabia	II	high	53
Medina (Yathrib)	iatrib	Arabia	II	high	65
Muscat	maskat	Arabia	II	high	338
Najran	negran	Arabia	II	high	135
Qatif	al katif	Arabia	III	high	236
Saada	saada	Arabia	I	high	134
Sanaa	sanaa	Arabia	I	high	101
Sohar	sohar	Arabia	II	high	339

Town	On the map	Region	Clim.	Reading	Local
Tarim	tarim	Arabia	I	high	314
Tayma	teima	Arabia	III	high	328
Zabid	zabid	Arabia	I	high	80
Zafar (Yemen)	tafar	Arabia	I	high	187
al-Ahsa (Hofuf)	al ahsa	Arabia	III	high	205
al-Juhfa	al gofa	Arabia	II	medium	89
al-Shihr	madina al sihr	Arabia	II	high	208
Akhsikath	ahsikat	Central Asia and Afghanistan	IV	high	58
Andarab	indarab	Central Asia and Afghanistan	III	high	250
Balkh	balh	Central Asia and Afghanistan	III	high	84
Bost (Lashkar Gah)	bost	Central Asia and Afghanistan	II	high	288
Bukhara	buhari	Central Asia and Afghanistan	IV	high	92
Hazarasp	hazarasb	Central Asia and Afghanistan	IV	high	163
Herat	herat	Central Asia and Afghanistan	III	medium	280
Jizzakh	gizak	Central Asia and Afghanistan	IV	medium	216
Kabul	kabul	Central Asia and Afghanistan	II	medium	78
Kandahar	al kandahar	Central Asia and Afghanistan	II	medium	586
Kath (Khwarazm)	kat	Central Asia and Afghanistan	IV	medium	47
Khujand	hogand	Central Asia and Afghanistan	IV	high	48
Kunya-Urgench (Gurganj)	gorgania	Central Asia and Afghanistan	IV	high	163
Merv	maru al sahgan	Central Asia and Afghanistan	IV	high	164
Paykand	baikand	Central Asia and Afghanistan	IV	high	24
Samarkand	samarkand	Central Asia and Afghanistan	IV	medium	48
Sarakhs	sarahs	Central Asia and Afghanistan	III	high	111
Zaranj	zarang	Central Asia and Afghanistan	III	high	210
Augsburg	auzburk	Central and Eastern Europe	VI	high	206
Basel	bazla	Central and Eastern Europe	VI	high	105
Belgrade	belgraba	Central and Eastern Europe	VI	high	521
Chersonesus (Kherson)	karsuna	Central and Eastern Europe	VI	medium	217
Cologne	kolunia	Central and Eastern Europe	VI	high	80
Gniezno	genasia	Central and Eastern Europe	VI	medium	566
Krakow	akrakal	Central and Eastern Europe	VI	medium	300
Kyiv	kiau	Central and Eastern Europe	VI	medium	657

Town	On the map	Region	Clim.	Reading	Local
Mainz	maiansa	Central and Eastern Europe	VI	high	164
Nitra	neitram	Central and Eastern Europe	VI	high	66
Akhmim	ahmim	Egypt and Nubia	II	high	70
Alexandria	al iskanderia	Egypt and Nubia	III	high	21
Antinoe (Ansina)	ansana	Egypt and Nubia	II	high	33
Armant	arment	Egypt and Nubia	II	high	111
Aswan	asuan	Egypt and Nubia	I	high	71
Asyut	assiut	Egypt and Nubia	II	high	208
Aydhab	aidab	Egypt and Nubia	II	high	161
El Arish	al aris	Egypt and Nubia	III	high	34
El Mahalla el Kubra	al mahalla	Egypt and Nubia	III	high	23
Esna	asnah	Egypt and Nubia	II	high	72
Fustat (Old Cairo)	misr	Egypt and Nubia	III	high	17
Heliopolis (Ain Shams)	ain sams	Egypt and Nubia	III	high	72
Hermopolis (al-Ashmunayn)	asmun	Egypt and Nubia	II	high	61
Ihnasya (Ahnas)	ahnas	Egypt and Nubia	II	high	64
Minya	miniat ibn al hasib	Egypt and Nubia	II	medium	56
Old Dongola	donkla	Egypt and Nubia	I	medium	657
Oxyrhynchus (al-Bahnasa)	al bahnasa	Egypt and Nubia	II	high	27
Pelusium (al-Farama)	al farama	Egypt and Nubia	III	high	61
Pharan (Feiran)	faran	Egypt and Nubia	III	medium	101
Qift	kaft	Egypt and Nubia	II	high	130
Qus	kus	Egypt and Nubia	II	high	87
Rosetta (Rashid)	rasid	Egypt and Nubia	III	high	12
Samannud	samnud	Egypt and Nubia	III	medium	11
Siwa	santaria	Egypt and Nubia	II	high	29
Suez (al-Qulzum)	al kolzum	Egypt and Nubia	III	high	60
Dover	dobrs	England	VI	high	45
Grimsby	agrims	England	VII	high	153
Hastings	hastinks	England	VI	high	10
Lincoln	nikula	England	VII	medium	115
London	londr	England	VI	high	55
Salisbury	salaberis	England	VII	high	170
Shoreham-by-Sea	sorham	England	VI	medium	21
Wareham	garham	England	VI	medium	71
Angouleme	ankolazma	France and the Low Countries	V	medium	24
Besancon	besanzun	France and the Low Countries	VI	high	187
Bordeaux	bordal	France and the Low Countries	V	high	79
Bruges	abrugs	France and the Low Countries	VI	high	35
Cahors	kaurs	France and the Low Countries	V	medium	160
Carcassonne	karkasuna	France and the Low Countries	IV	high	55
Chartres	gartas	France and the Low Countries	VI	high	43
Dijon	digun	France and the Low Countries	V	high	73
Dinan	dinam	France and the Low Countries	VI	high	52
Dol-de-Bretagne	dul	France and the Low Countries	VI	high	43
Evreux	abraus	France and the Low Countries	VI	medium	59

Town	On the map	Region	Clim.	Reading	Local
Ghent	kent	France and the Low Countries	VI	high	71
Le Mans	al mais	France and the Low Countries	VI	medium	81
Liege	liaga	France and the Low Countries	VI	high	29
Limoges	limuges	France and the Low Countries	V	high	129
Lyon	liun	France and the Low Countries	V	high	78
Marseille	masilia	France and the Low Countries	V	high	69
Nantes	nanitas	France and the Low Countries	V	high	70
Narbonne	arbuna	France and the Low Countries	IV	high	97
Nevers	nifaras	France and the Low Countries	V	medium	53
Orleans	arlians	France and the Low Countries	VI	high	28
Paris	abariz	France and the Low Countries	VI	high	10
Poitiers	bataras	France and the Low Countries	V	medium	83
Quimper (Saint-Corentin)	kerenbin sant karnin	France and the Low Countries	VI	medium	8
Rouen	ratumagus	France and the Low Countries	VI	high	53
Saint-Gilles	sant gili	France and the Low Countries	V	high	30
Saint-Malo	sant mahlu	France and the Low Countries	VI	high	54
Saint-Valery-sur-Somme	sant ualeri	France and the Low Countries	VI	medium	17
Toulouse	tolusa	France and the Low Countries	V	high	85
Tours	tars	France and the Low Countries	VI	medium	80
Utrecht	astrik	France and the Low Countries	VI	medium	16
Verdun	berdun	France and the Low Countries	VI	high	77
Abydos (Dardanelles)	abidus	Greece and the Balkans	V	high	17
Ahtopol (Agathopolis)	agatubolus	Greece and the Balkans	VI	high	50
Argos	argu	Greece and the Balkans	IV	high	22
Athens	atina	Greece and the Balkans	V	high	156
Bar (Antivari)	antibra	Greece and the Balkans	V	high	105
Constantinople (Istanbul)	al kostantina	Greece and the Balkans	VI	high	119
Corfu	marnisa (kurfus)	Greece and the Balkans	IV	high	187
Corinth	korint	Greece and the Balkans	IV	high	29
Demetrias	dimitriana	Greece and the Balkans	V	medium	44
Dubrovnik (Ragusa)	ragusa	Greece and the Balkans	V	high	86
Edirne (Adrianople)	adirnuboli	Greece and the Balkans	V	high	77
Gallipoli (Gelibolu)	kalibolu	Greece and the Balkans	V	high	45
Kitros (Pydna)	kitrus	Greece and the Balkans	V	medium	111
Larissa	larisa	Greece and the Balkans	V	medium	147
Luleburgaz (Arcadiopolis)	arkadiuboli	Greece and the Balkans	V	high	16
Marmara Ereglisi (Heraclea)	heraklia	Greece and the Balkans	V	high	20
Monemvasia	maliasa	Greece and the Balkans	IV	high	83
Nafplio	anubali	Greece and the Balkans	IV	high	27

Town	On the map	Region	Clim.	Reading	Local
Nesebar (Mesembria)	mesinus	Greece and the Balkans	VI	medium	84
Ohrid	ahrida	Greece and the Balkans	V	high	40
Patras	bras	Greece and the Balkans	IV	medium	158
Plovdiv (Philippopolis)	filibus	Greece and the Balkans	VI	high	167
Preslav	barsklafa	Greece and the Balkans	VI	medium	78
Silistra (Dristra)	darsira	Greece and the Balkans	VI	medium	147
Silivri (Selymbria)	salimria	Greece and the Balkans	V	high	40
Sozopol	sizuboli	Greece and the Balkans	VI	high	42
Tekirdag (Rodosto)	rudostu	Greece and the Balkans	V	high	28
Thebes	astifas	Greece and the Balkans	V	high	69
Thessaloniki	salunik	Greece and the Balkans	V	high	141
Varna	barnas	Greece and the Balkans	VI	high	168
Vize	bizi	Greece and the Balkans	VI	high	55
Zadar	gadra	Greece and the Balkans	V	medium	76
Alcacer do Sal	al kasr	Iberia	IV	high	105
Alcantara	al kantara al sif	Iberia	IV	high	36
Algeciras	al gezira	Iberia	IV	high	26
Alicante	lekant	Iberia	IV	high	39
Almeria	al meria	Iberia	IV	high	37
Almunecar	al monkab	Iberia	IV	high	21
Archidona	arsiduna	Iberia	IV	high	3
Astorga	astorka	Iberia	V	high	17
Badajoz	batalius	Iberia	IV	high	73
Baeza	baisa	Iberia	IV	high	53
Baza	basta	Iberia	IV	high	31
Burgos	burgas	Iberia	V	high	36
Burriana	boriana	Iberia	IV	high	27
Calatayud	kalaat aiub	Iberia	IV	high	55
Calatrava la Vieja	kalaat rabah	Iberia	IV	high	36
Cartagena	kartagena	Iberia	IV	high	77
Coimbra	kalimria	Iberia	IV	high	67
Cordoba	kortoba	Iberia	IV	medium	22
Coria	koria	Iberia	IV	high	46
Cuenca	kuenka	Iberia	IV	high	65
Denia	dania	Iberia	IV	high	11
Ecija	asiha	Iberia	IV	high	44
Elche	els	Iberia	IV	high	8
Evora	iabara	Iberia	IV	high	17
Fraga	afraga	Iberia	IV	high	32
Granada	al granata	Iberia	IV	high	8
Guadalajara	uadi al hagara	Iberia	IV	medium	40
Guadix	uadias	Iberia	IV	high	14
Huesca	ueska	Iberia	IV	high	92
Ibiza	iabesa	Iberia	IV	high	52
Jaen	gijan	Iberia	IV	high	49
Jerez de la Frontera	seris	Iberia	IV	high	25
Leon	liun	Iberia	V	high	71
Lisbon	asbuna	Iberia	IV	high	20
Lleida	lerida	Iberia	IV	high	69
Logrono	kastila (logrono)	Iberia	V	medium	56
Lorca	lurka	Iberia	IV	high	56
Majorca (Palma)	maiorka	Iberia	IV	high	79
Malaga	malaka	Iberia	IV	medium	36
Medellin	medellin	Iberia	IV	high	44
Medinaceli	medinat salim	Iberia	IV	high	27
Merida	merida	Iberia	IV	medium	71
Mertola	martala	Iberia	IV	high	70
Murcia	mursia	Iberia	IV	high	28
Najera	nagra	Iberia	V	high	30
Niebla	libla	Iberia	IV	high	19

Town	On the map	Region	Clim.	Reading	Local
Osuna	osuna	Iberia	IV	high	39
Pamplona	banbluna	Iberia	V	high	47
Santarem	santarin	Iberia	IV	high	85
Santiago de Compostela	sant iakub	Iberia	V	high	26
Seville	isbilia	Iberia	IV	high	34
Silves	silb	Iberia	IV	medium	95
Talavera de la Reina	talabera	Iberia	IV	high	87
Tarifa	tarif	Iberia	IV	high	23
Toledo	tolaitola	Iberia	IV	high	28
Tortosa	tartusa	Iberia	IV	high	37
Tudela	tutila	Iberia	IV	high	86
Tui	duia	Iberia	V	medium	68
Ubeda	abda	Iberia	IV	high	34
Ucles	uklis	Iberia	IV	high	38
Valencia	balinsia	Iberia	IV	high	31
Xativa	satiba	Iberia	IV	high	74
Zamora	samura	Iberia	V	high	104
Zaragoza	sarakostia	Iberia	IV	high	30
Abhar	abhar	Iran and the Caucasus	IV	high	41
Ahvaz	al ahuaz	Iran and the Caucasus	III	high	81
Ardabil	ardabil	Iran and the Caucasus	V	high	177
Barda	bardaa	Iran and the Caucasus	V	high	93
Bastam	bastam	Iran and the Caucasus	IV	high	14
Derbent	bab al abuab	Iran and the Caucasus	V	high	305
Dinawar	dainur	Iran and the Caucasus	IV	high	76
Dvin	dabil	Iran and the Caucasus	V	high	106
Fasa	fasa	Iran and the Caucasus	III	high	285
Gorgan (Astarabad)	asterabad	Iran and the Caucasus	IV	high	210
Hamadan	hamadan	Iran and the Caucasus	IV	high	149
Isfahan	isbahan	Iran and the Caucasus	III	high	36
Istakhr	istahr	Iran and the Caucasus	III	high	47
Jiroft	giraft	Iran and the Caucasus	III	high	47
Kashan	kasan	Iran and the Caucasus	IV	high	38
Kazerun	kazerun	Iran and the Caucasus	III	high	92
Kermanshah (Qirmisin)	karmasin	Iran and the Caucasus	IV	high	27
Khoy	hui	Iran and the Caucasus	V	high	105
Maragheh	al maraga	Iran and the Caucasus	IV	high	39
Nahavand	nahauand	Iran and the Caucasus	IV	high	4
Qayen	kain	Iran and the Caucasus	III	high	111
Qom	kum	Iran and the Caucasus	IV	high	97
Rayy	al rai	Iran and the Caucasus	IV	high	54
Salmas	salmas	Iran and the Caucasus	V	high	111
Shamakhi	al samahia	Iran and the Caucasus	V	high	160
Shiraz	siraz	Iran and the Caucasus	III	high	30
Sirjan	al sirgan	Iran and the Caucasus	III	high	58
Susa (Shush)	sus	Iran and the Caucasus	III	high	20
Tabas	al tabein	Iran and the Caucasus	III	medium	206
Tabriz	tabriz	Iran and the Caucasus	IV	high	141
Tbilisi	tiflis	Iran and the Caucasus	V	high	81
Tiz	al taiza	Iran and the Caucasus	II	medium	116
Yazd	iazd	Iran and the Caucasus	III	high	37
Zanjan	zangan	Iran and the Caucasus	IV	high	65
Abadan	abadan	Iraq and the Jazira	III	high	49
Baghdad	bagdad	Iraq and the Jazira	IV	high	27
Basra	al basra	Iraq and the Jazira	III	high	62
Circesium (Qarqisiya)	kirkisia	Iraq and the Jazira	IV	high	117
Ctesiphon (al-Madain)	al madain	Iraq and the Jazira	IV	medium	54
Daskara	al daskara	Iraq and the Jazira	IV	high	16
Diyarbakir (Amid)	amid	Iraq and the Jazira	IV	high	59
Edessa (Urfa)	al roha	Iraq and the Jazira	IV	high	36

Town	On the map	Region	Clim.	Reading	Local
Haditha (on the Tigris)	al hadita	Iraq and the Jazira	IV	medium	128
Harran	harran	Iraq and the Jazira	IV	high	46
Hit	hit	Iraq and the Jazira	IV	high	57
Karbala	kerbela	Iraq and the Jazira	IV	high	86
Khanaqin	hanikin	Iraq and the Jazira	IV	high	39
Kufa	al kufa	Iraq and the Jazira	IV	high	76
Mosul	al mosul	Iraq and the Jazira	IV	high	86
Raqqa	al rakka	Iraq and the Jazira	IV	medium	49
Samarra	sorra men ra	Iraq and the Jazira	IV	high	41
Silvan (Mayyafariqin)	miafark	Iraq and the Jazira	V	high	65
Tikrit	tekrit	Iraq and the Jazira	IV	high	107
Ukbara	okbara	Iraq and the Jazira	IV	high	9
al-Anbar	al anbar	Iraq and the Jazira	IV	high	12
al-Qadisiyya	al kadesia	Iraq and the Jazira	III	high	54
al-Rahba (Mayadin)	rahbat malik	Iraq and the Jazira	IV	high	7
Amalfi	malf	Italy and Corsica	V	high	30
Ancona	ankuna	Italy and Corsica	V	medium	38
Bari	bari	Italy and Corsica	V	high	76
Benevento	benebent	Italy and Corsica	V	high	11
Brindisi	abrindis	Italy and Corsica	V	high	35
Capua	kabua	Italy and Corsica	V	high	39
Corsica	korkisa	Italy and Corsica	IV	medium	36
Fano	fanu	Italy and Corsica	V	high	36
Florence	fluranaia	Italy and Corsica	V	medium	95
Gaeta	gabata	Italy and Corsica	V	high	28
Genoa	ganua	Italy and Corsica	V	high	81
Luni	luna	Italy and Corsica	V	medium	97
Mantua	mantua	Italy and Corsica	V	high	40
Monopoli	manubli	Italy and Corsica	V	high	38
Naples	nabal	Italy and Corsica	V	high	14
Pisa	bisa	Italy and Corsica	V	high	49
Ravenna	rabena	Italy and Corsica	V	high	20
Rimini	ariminus	Italy and Corsica	V	high	25
Rome	ruma	Italy and Corsica	V	high	176
Salerno	salarnu	Italy and Corsica	V	high	39
Sorrento	sarant	Italy and Corsica	V	high	26
Taranto	tarant	Italy and Corsica	V	high	59
Terracina	targina	Italy and Corsica	V	high	37
Verona	beruna	Italy and Corsica	V	medium	58
Aghmat	agmat uarika	Morocco and Algeria	III	high	70
Algiers	gezair bani mazgani	Morocco and Algeria	III	high	32
Anfa (Casablanca)	anfa	Morocco and Algeria	III	high	27
Annaba (Bone)	buna	Morocco and Algeria	III	high	11
Badis	badis	Morocco and Algeria	III	medium	30
Bejaia	begaia	Morocco and Algeria	III	high	6
Biskra	biskra	Morocco and Algeria	III	high	87
Ceuta	sebta	Morocco and Algeria	IV	high	47
Cherchell	sersal	Morocco and Algeria	III	high	71
Collo	al kol	Morocco and Algeria	III	high	36
Constantine	kosantina	Morocco and Algeria	III	high	34
Dellys	tadellis	Morocco and Algeria	III	high	17
Fez	fas	Morocco and Algeria	III	high	17
Ksar el-Kebir	kasr abd al karim	Morocco and Algeria	III	high	39
M'Sila	al mesila	Morocco and Algeria	III	high	52
Marrakesh	marrakes	Morocco and Algeria	III	high	76
Meknes	miknasa	Morocco and Algeria	III	high	65
Melilla	melilla	Morocco and Algeria	III	high	21
Mila	mila	Morocco and Algeria	III	high	25
Miliana	meliana	Morocco and Algeria	III	high	68
Oran	uahran	Morocco and Algeria	III	medium	114

Town	On the map	Region	Clim.	Reading	Local
Qal'at Beni Hammad	kalaat bani hamud	Morocco and Algeria	III	high	17
Safi	asafi	Morocco and Algeria	III	high	117
Sale	sala	Morocco and Algeria	III	high	24
Setif	setif	Morocco and Algeria	III	high	53
Sijilmasa	sigilmasa	Morocco and Algeria	III	high	385
Tahert (Tiaret)	tahort	Morocco and Algeria	III	high	37
Tangier	tanga	Morocco and Algeria	IV	high	22
Taroudant	tarudant	Morocco and Algeria	III	high	92
Tetouan	tetauin	Morocco and Algeria	IV	high	33
Tinmel	tinmalal	Morocco and Algeria	III	high	167
Tlemcen	talamsan	Morocco and Algeria	III	high	177
Tobna	tobna	Morocco and Algeria	III	high	29
Agrigento	karkant	Sicily and Malta	IV	medium	11
Butera	butira	Sicily and Malta	IV	high	42
Caronia	karunia	Sicily and Malta	IV	high	20
Cefalu	kalfudi	Sicily and Malta	IV	high	6
Enna (Castrogiovanni)	kasr iani	Sicily and Malta	IV	high	25
Lentini	lentini	Sicily and Malta	IV	high	21
Malta	melita	Sicily and Malta	IV	high	78
Mazara del Vallo	mazar	Sicily and Malta	IV	high	6
Messina	mesina	Sicily and Malta	IV	high	34
Milazzo	milas	Sicily and Malta	IV	high	5
Noto	nutas	Sicily and Malta	IV	high	61
Palermo	balarm	Sicily and Malta	IV	high	15
Patti	bagtus	Sicily and Malta	IV	high	16
Ragusa	ragus	Sicily and Malta	IV	high	9
Sciacca	al saka	Sicily and Malta	IV	medium	6
Scicli	sikla	Sicily and Malta	IV	high	21
Syracuse	sirakusa	Sicily and Malta	IV	high	34
Taormina	tarmi	Sicily and Malta	IV	high	21
Termini Imerese	tirma	Sicily and Malta	IV	high	26
Trapani	trabana	Sicily and Malta	IV	high	24
Chaul (Saymur)	saimur	Sind and India	II	medium	246
Daybul (Banbhore)	daibal	Sind and India	II	high	560
Kannauj	al kanug	Sind and India	II	high	767
Khambhat (Cambay)	kanbaia	Sind and India	II	high	247
Multan	al multan	Sind and India	II	high	385
Patan (Anhilwara)	nahruara	Sind and India	II	high	464
Sanjan	sandan	Sind and India	II	high	30
Sopara	subara	Sind and India	II	high	217
al-Mansura (Sindh)	al mansuria	Sind and India	II	high	215
Acre	akka	Syria and Palestine	III	high	19
Aleppo	haleb	Syria and Palestine	IV	medium	26
Antioch (Antakya)	antakiu	Syria and Palestine	IV	high	47
Aqaba (Ayla)	aila	Syria and Palestine	III	medium	77
Arqa	arka	Syria and Palestine	III	high	21
Ashkelon	askalan	Syria and Palestine	III	high	24
Baalbek	baalbek	Syria and Palestine	III	high	51
Balis	balis	Syria and Palestine	IV	high	45
Beirut	beirut	Syria and Palestine	III	high	25
Byblos (Jbeil)	gebail	Syria and Palestine	III	high	6
Caesarea	kaisaria	Syria and Palestine	III	high	8
Damascus	dimask	Syria and Palestine	III	high	40
Daraa (Adhriat)	adraat	Syria and Palestine	III	high	49
Gaza	gazza	Syria and Palestine	III	high	13
Haifa	haifa	Syria and Palestine	III	high	23
Hebron	al halil	Syria and Palestine	III	high	57
Homs	homs	Syria and Palestine	III	high	40
Jaffa	iafa	Syria and Palestine	III	high	12
Jericho	eriha	Syria and Palestine	III	high	14

Town	On the map	Region	Clim.	Reading	Local
Jerusalem	al kuds	Syria and Palestine	III	high	44
Latakia	al ladikija	Syria and Palestine	IV	high	41
Manbij	manbig	Syria and Palestine	IV	high	62
Nicosia	lafkisia	Syria and Palestine	IV	medium	68
Palmyra (Tadmur)	tadmir	Syria and Palestine	III	high	64
Qinnasrin	kinnesrin	Syria and Palestine	IV	high	22
Rafah	refah	Syria and Palestine	III	high	6
Ramla	al ramla	Syria and Palestine	III	high	3
Salamiyah	salamia	Syria and Palestine	IV	high	64
Sidon	saida	Syria and Palestine	III	high	7
Tartus	antarsus	Syria and Palestine	IV	high	26
Tiberias	tabaria	Syria and Palestine	III	high	4
Tripoli (Lebanon)	atrablus	Syria and Palestine	III	high	36
Tyre	sur	Syria and Palestine	III	high	16
Ajdabiya	agdabia	Tunisia and Libya	III	high	76
Awjila	augila	Tunisia and Libya	III	high	212
Barca (al-Marj)	barka	Tunisia and Libya	III	high	135
Beja	baga	Tunisia and Libya	III	high	9
Carthage	kartagina	Tunisia and Libya	III	high	46
Gabes	kafs	Tunisia and Libya	III	medium	84
Gafsa	kafsa	Tunisia and Libya	III	high	36
Ghadames	gadames	Tunisia and Libya	III	high	155
Kairouan	al kairuan	Tunisia and Libya	III	high	51
Lebda (Leptis Magna)	lebda	Tunisia and Libya	III	high	97
Mahdia	al mahdia	Tunisia and Libya	III	medium	20
Ptolemais (Tolmeita)	tolmaita	Tunisia and Libya	III	high	22
Sfax	asfakis	Tunisia and Libya	III	high	46
Sirte	sort	Tunisia and Libya	III	high	35
Sousse	susa	Tunisia and Libya	III	medium	16
Tabarka	tabarka	Tunisia and Libya	III	high	14
Tocra	teukara	Tunisia and Libya	III	high	143
Tozeur	tauzer	Tunisia and Libya	III	high	49
Tripoli (Libya)	atrablus	Tunisia and Libya	III	high	80
Tunis	tunis	Tunisia and Libya	III	high	54
Koumbi Saleh (capital of Ghana)	gana	West Africa	I	medium	664